

OpenVAS Vulnerability Assessment Report

Cybersecurity Internship – Task 3

Prepared By

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Tool Used

OpenVAS (Greenbone Vulnerability Manager)

Target System

Metasploitable2 Virtual Machine

Target IP Address

192.168.56.103

Scan Date

24 June 2026

1. Objective

The objective of this assessment was to perform a vulnerability scan on the Metasploitable2 virtual machine using OpenVAS and identify security weaknesses, vulnerable services, misconfigurations, and potential attack vectors.

2. Environment Setup

Component	Details
Scanner Machine	Kali Linux
Vulnerability Scanner	OpenVAS / Greenbone
Target Machine	Metasploitable2
Target IP	192.168.56.103

Component	Details
Scan Profile	Full and Fast

3. Scan Configuration

The OpenVAS scanner was configured using the default "Full and Fast" scan profile. The target host was added manually and scanned using the OpenVAS Default Scanner.

Configuration Details

- Scanner: OpenVAS Default
- Scan Profile: Full and Fast
- Target Host: 192.168.56.103
- Feed Status: Current
- NVT Feed: Updated
- SCAP Feed: Updated
- CERT Feed: Updated

4. Scan Results

The vulnerability assessment completed successfully and generated multiple findings.

Summary

Metric	Value
Hosts Scanned	1
Open Ports Detected	19
Applications Detected	19
Operating Systems Detected	1
CVEs Identified	34
Total Results	68
Maximum Severity	10.0 (Critical)

5. Vulnerability Statistics

The following vulnerabilities were identified during the assessment:

Severity	Count
Critical	12
High	8
Medium	32
Low	5
Informational	65

6. Critical Vulnerabilities Identified

6.1 Possible Backdoor: Ingreslock

Severity

Critical (CVSS 10.0)

Description

A possible backdoor service was detected on the target system. Attackers may exploit this service to gain unauthorized access.

Impact

- Unauthorized access
- Remote compromise
- System takeover

Recommendation

- Disable unnecessary services
 - Restrict network exposure
 - Apply security updates
-

6.2 rlogin Passwordless Login

Severity

Critical (CVSS 10.0)

Description

The rlogin service permits passwordless authentication.

Impact

- Authentication bypass
- Unauthorized remote access

Recommendation

- Disable rlogin service
 - Replace with SSH
-

6.3 rexec Service Running

Severity

Critical (CVSS 10.0)

Description

The rexec service allows remote command execution and transmits credentials insecurely.

Impact

- Remote code execution
- Credential exposure

Recommendation

- Disable rexec service
 - Use encrypted remote administration protocols
-

6.4 Distributed Ruby (dRuby) Multiple RCE Vulnerabilities

Severity

Critical (CVSS 10.0)

Description

Multiple vulnerabilities in Distributed Ruby may allow remote code execution.

Impact

- Arbitrary code execution
- Full system compromise

Recommendation

- Update affected software
 - Restrict network access
-

6.5 Operating System End-of-Life Detection

Severity

Critical (CVSS 10.0)

Description

The detected operating system is no longer supported by the vendor.

Impact

- Unpatched vulnerabilities
- Increased attack surface

Recommendation

- Upgrade to a supported operating system
-

6.6 Apache Tomcat AJP Remote Code Execution (Ghostcat)

Severity

Critical (CVSS 9.8)

Description

Apache Tomcat is vulnerable to the Ghostcat vulnerability which may allow remote code execution.

Impact

- File disclosure
- Remote code execution

Recommendation

- Upgrade Apache Tomcat
 - Restrict AJP connector access
-

6.7 MySQL/MariaDB Default Credentials

Severity

Critical (CVSS 9.8)

Description

Default database credentials were detected.

Impact

- Database compromise
- Data theft

Recommendation

- Change default credentials
 - Enforce strong passwords
-

6.8 vsFTPD Backdoor Vulnerability

Severity

Critical (CVSS 9.8)

Description

The FTP service is running a backdoored version of vsFTPD.

Impact

- Remote shell access
- Full system compromise

Recommendation

- Upgrade FTP software
 - Remove vulnerable versions
-

6.9 DistCC Remote Code Execution

Severity

Critical (CVSS 9.3)

Description

The DistCC service allows arbitrary command execution.

Impact

- Remote code execution
- Host compromise

Recommendation

- Disable DistCC if unused
 - Restrict network access
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7. Risk Assessment

The vulnerability scan revealed multiple critical security weaknesses affecting remote services, authentication mechanisms, operating system support status, and exposed applications.

Several findings allow:

- Remote Code Execution (RCE)
- Authentication Bypass
- Unauthorized Remote Access
- Privilege Escalation
- Data Exposure

The target system presents a very high attack surface and would be considered highly vulnerable in a production environment.

8. Recommendations

The following actions are recommended:

1. Disable insecure services such as Telnet, rlogin, and rexec.
 2. Remove or update vulnerable software versions.
 3. Change all default credentials.
 4. Restrict access to administrative services.
 5. Apply operating system and application patches.
 6. Implement firewall rules to reduce attack surface.
 7. Perform regular vulnerability assessments.
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9. Conclusion

An OpenVAS vulnerability assessment was successfully performed against the Metasploitable2 target system. The scan identified 122 vulnerabilities, including 12 Critical findings and multiple High and Medium severity issues. Several vulnerabilities allow remote code execution, authentication bypass, and unauthorized access. Appropriate remediation measures should be implemented to improve the security posture of the system.

The assessment demonstrates the effectiveness of OpenVAS in identifying security weaknesses and highlights the importance of vulnerability management in maintaining secure systems.